MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSEAQC201 – AQUATIC ECOLOGY, WATER AND SOIL QUALITY PARAMETERS

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer All Questions
Each question carries 1 mark

1.What does the term Benthos refer to? [1] [U]

2. Name any two chemical water quality parameters? [2,6] [U]

3.What is the unit for expressing the level of Ammonia in water? [2,6] [U]

4. Define microplastics. [5] [U]

5. Define salinity. [2,6] [U]

6. What are zooplankton? [3] [U]

7. What is the purpose of a Secchi disc in water quality assessment? [4,6] [U]

8. List the different zones of a lake based on light penetration? [1] [U]

9. Define bio-indicators? [3] [U]

10. What is haphazard sampling? [4] [U]

[1 x 10 =10] Marks

Part B
Short Answer Questions
Answer any 5 Questions
Each question carries 6 marks

13. Explain the importance of temperature, salinity and turbidity as physical quality parameters. [2,6] [A]

14. Discuss the ecological significance of phytoplankton and zooplankton in water quality monitoring. [3] [A]

15. Explain the ecological impacts of turbidity. [2,6] [U]

16. What are harmful algal blooms? What causes HABs and what are their impacts on aquatic ecosystems? [5] [U]

17. What are the adaptations of lotic animals? [1] [A]

18. Explain the different methods of soil sampling used in aquaculture. [4] [U]

19. Describe the process of soil formation and explain its composition. [1] [U]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer any one Question
Each question carries 10 marks

20. Discuss the impact of pharmaceutical contaminants and emerging pathogens on aquatic ecosystems and aquaculture practices. [5] [A]

21. How are aquatic micro and macroinvertebrates used as bio-indicators of water quality? Explain with examples and their importance in monitoring aquatic environments. [3] [A]

22. What are the positive and negative impacts of dams on the environment, aquatic life and nearby communities? Explain with examples. [1] [A]

[10 x 1 = 10] Marks